MODULE 4: SPECTROSCOPIC AND DIFFRACTION TECHNIQUES

(CO: 3)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 6 Topics): (1) Fundamental concepts in spectroscopic techniques, (2) Fundamental concepts in diffraction techniques, (3) Principle and instrumentation of UV-Visible spectroscopy, (4) Applications of UV-Visible spectroscopy, (5) Powder XRD: Determination of crystal size numericals - Part 1, and (6) Powder XRD: Determination of crystal size numericals - Part 2.

1 VIT FAT Exam Blueprint & Master Formula Sheet

Module 4 carries **15 to 20 marks** in BACHY101 FAT:

1. **Question Type A (10 Marks):** Beer-Lambert's Law derivation, electronic transitions ($\sigma \to \sigma^*, n \to \sigma^*, \pi \to \pi^*, n \to \pi^*$), chromophores, auxochromes, and shifts (bathochromic vs hypsochromic).
2. **Question Type B (10 Marks):** Powder X-Ray Diffraction (XRD): Bragg's Law derivation ($2d \sin\theta = n\lambda$), Scherrer equation derivation ($D = \frac{K\lambda}{\beta \cos\theta}$) and worked numericals to compute average crystallite size from FWHM.

Technique / Law	Governing Equation / Formula	VIT Exam Key Insight
Beer-Lambert Law	$A = \log_{10}\left(\frac{I_0}{I}\right) = \varepsilon c l$	Absorbance A is dimensionless; ε is molar absorptivity ($\frac{\text{L}}{\text{mol} \cdot \text{cm}}$).
Bragg's Law of Diffraction	$2d_{hkl} \sin\theta = n\lambda$	$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$ for cubic crystal lattices.
Debye-Scherrer Equation	$D = \frac{K\lambda}{\beta \cos\theta}$	$K = 0.9$ (shape factor), β is FWHM in RADIANS, not degrees!
FWHM Unit Conversion	$\beta(\text{rad}) = \beta(\text{degrees}) \times \frac{\pi}{180}$	Most common VIT FAT student error is omitting radian conversion.

2 UV-Visible Spectroscopy

2.1 Electromagnetic Spectrum & Electronic Transitions

Spectroscopy is the study of interaction between electromagnetic radiation and matter. In the UV-Visible region ($200 - 800$ nm), photon absorption induces promotion of valence electrons from bonding or non-bonding orbitals to anti-bonding molecular orbitals:

- $\sigma \to \sigma^*$ transition: High energy, deep vacuum UV (< 180 nm). Example: Methane (CH_4).
- $n \to \sigma^*$ transition: Non-bonding electron on heteroatom (O, N, S, halogens) to σ^*. Wavelength $180 - 250$ nm. Example: Water, alcohols, amines.
- $\pi \to \pi^*$ transition: Found in unsaturated hydrocarbons with double/triple bonds. Wavelength $200 - 400$ nm. High molar absorptivity ($\varepsilon > 10^4$).
- $n \to \pi^*$ transition: Promotion of non-bonding lone pair to π^* antibonding orbital. Wavelength $270 - 350$ nm. Weak intensity ($\varepsilon \approx 10 - 100$) because it is symmetry-forbidden. Example: Carbonyls ($> C = O$).

2.2 Chromophores, Auxochromes & Spectral Shifts

- **Chromophore:** An isolated covalently bonded unsaturated group responsible for electronic absorption in the UV-Vis range (e.g., $-C = C -, -C = O, -N = N -, -NO_2$).

- **Auxochrome:** A saturated substituent group with non-bonding lone pairs that does not absorb UV light on its own, but when attached to a chromophore, alters both the wavelength and intensity of absorption (e.g., $-OH, -NH_2, -SH, -Cl$).
- **Bathochromic Shift (Red Shift):** Absorption maximum shifts to a longer wavelength (lower energy), frequently caused by conjugation or auxochrome substitution.
- **Hypsochromic Shift (Blue Shift):** Shift of λ_{max} to shorter wavelength (higher energy), caused by loss of conjugation or protonation.
- **Hyperchromic Effect:** An increase in molar absorption intensity (ε).
- **Hypochromic Effect:** A decrease in molar absorption intensity (ε).

2.3 Instrumentation: Double-Beam UV-Vis Spectrophotometer

Components: Deuterium lamp (UV: $190-380$ nm) and Tungsten-halogen lamp (Vis: $380-800$ nm) $\rightarrow$ Monochromator (diffraction grating) $\rightarrow$ Beam splitter / Rotating chopper (divides beam into sample beam and reference beam) $\rightarrow$ Sample and Reference quartz cuvettes $\rightarrow$ Photomultiplier tube (PMT) detector $\rightarrow$ Signal processor and recorder.

3 Powder X-Ray Diffraction (Powder XRD)

3.1 Principle: Bragg's Law

When monochromatic X-rays of wavelength λ strike parallel crystal lattice planes separated by interplanar spacing d_{hkl} at an angle θ, constructive interference occurs when the path difference between scattered rays from adjacent planes is an integral multiple of λ:

$$2d \sin\theta = n\lambda$$

For a cubic crystal system with lattice parameter a and Miller indices (h, k, l):

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

3.2 Crystallite Size Determination: Scherrer Formula

In macroscopic single crystals, diffraction peaks are infinitesimally sharp delta functions. When crystallites are smaller than 100 nm, incomplete destructive interference broadens the diffraction peaks. The average crystallite size D is quantified by the **Debye-Scherrer Equation**:

$$D = \frac{K\lambda}{\beta \cos\theta}$$

where:

- K is the dimensionless crystallite shape factor (typically 0.9 or 0.94).
- λ is the X-ray wavelength (typically 0.15406 nm for $Cu-K_\alpha$ radiation).
- β is the Full Width at Half Maximum (FWHM) of the diffraction peak in **radians**:

$$\beta = \beta_{\text{observed}} - \beta_{\text{instrumental}}$$

- θ is the Bragg diffraction angle (half of the 2θ peak position).

4 Worked VIT Question Bank & Past FAT Problems

Problem 4.1 (VIT Question Bank Q16 & FAT Classic - 10 Marks): In a powder XRD experiment using $Cu-K_\alpha$ radiation ($\lambda = 0.15418$ nm), a nanomaterial shows a dominant diffraction peak at $2\theta = 38.2°$. The Full Width at Half Maximum (FWHM) of the peak is measured to be $0.48°$. Assuming the Scherrer shape constant $K = 0.90$, calculate the average crystallite size D in nanometers.

Solution:

1. Identify Bragg angle θ:

$$2\theta = 38.2° \Rightarrow \theta = \frac{38.2}{2} = 19.1°$$

$$\cos\theta = \cos(19.1^\circ) = 0.9449$$

2. Convert FWHM β from degrees to radians (CRITICAL STEP):

$$\beta = 0.48^\circ \times \frac{\pi}{180} = 0.48 \times 0.017453 = 8.3776 \times 10^{-3} \text{ radians}$$

3. Substitute into the Debye-Scherrer equation:

$$D = \frac{K\lambda}{\beta \cos\theta} = \frac{0.90 \times 0.15418 \times 10^{-9} \text{ m}}{8.3776 \times 10^{-3} \times 0.9449}$$

$$D = \frac{0.13876 \times 10^{-9}}{7.9160 \times 10^{-3}} = 1.753 \times 10^{-8} \text{ m} = 17.53 \text{ nm}$$

Answer: The average crystallite size is **17.53 nm**.